5CS037

Concepts

and

Technologies of AI

Predicting the Composite SDG Index Score of Nations from UN Sustainable Development Goal Indicator Scores (2000–2022)

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Abstract

Purpose: This report predicts a continuous target variable the overall SDG Index Score of a country in a given year using regression techniques, as part of the end-to-end machine learning workflow required for the 5CS037 Final Portfolio.

Dataset: The dataset that was used for this research is Sustainable Development Report/SDG Index Panel Data (2000-2022), which has 4,140 observations (180 countries × up to 23 years) and consists of 21 variables, such as the aggregate sdg_index_score and 17 sub-scores for each of 17 UNSDGs. This dataset is consistent with the UN Sustainable Development Goals (UNSDGs), because it is actually the tool of their monitoring and ranking.

Approach: The methodology involves EDA, creation of a Multi-Layer Perceptron (MLP) regressor neural network, two classic baseline models (Linear Regression and Random Forest Regressor), tuning hyperparameters through GridSearchCV/RandomizedSearchCV, selection of features via filter (SelectKBest) and embedded (Random Forest) methods, and finally a comparison of the tuned & selected features-based models.

Main Findings: Models were assessed by the metrics of MAE, MSE, RMSE, and R^2. The baseline Random Forest Regressor had the best overall test results (RMSE = 0.5340, R^2 = 0.9977) with a slight margin over the Neural Network (RMSE = 0.6885, R^2 = 0.9961) and quite significant margin over Linear Regression (RMSE = 1.0585, R^2 = 0.9909). In terms of the best tuned and selected features model, Random Forest (RMSE = 0.7249, R^2 = 0.995

Conclusion: Random Forest Regressor algorithm emerges as the most robust model for the task on all experiment settings. A significant reduction of the feature set to 8 out of 19 proved to affect negatively the performance of the linear regression model, thus demonstrating the necessity of comparison of multiple models' variations including baseline and tuned ones.

Contents

1.Introduction

1.1 Problem Statement

The goal of this project is to predict a continuous target variable — the composite SDG Index Score assigned to a country in a given year — from that country's 17 individual UN Sustainable Development Goal (SDG) sub-scores and basic panel identifiers (year, country). This is framed as a supervised regression problem: given a vector of goal-level scores, estimate the single number that summarises a country's overall sustainable-development performance.

1.2 Dataset

The dataset employed in the present analysis is called the Sustainable Development Report SDG Index panel (filename: sdg_index_2000-2022.csv). This dataset comes from the Sustainable Development Solutions Network (SDSN), an organization responsible for the publication of the annual Sustainable Development Report (previously the SDG Index and Dashboards) since 2016, under the supervision of Jeffrey D. Sachs and coauthors. The SDG Index and Dashboards were initiated back in 2016 by the SDSN and Bertelsmann Stiftung. Moreover, its methodology has been peer-reviewed (for example, Nature Geoscience) and audited by the Joint Research Centre of the European Commission. The utilized data represents an extract of historical panel data (from 2000 to 2022) of the country-level scores provided by the SDSN through their website SDG Transformation Center / sdgindex.org.

This data set addresses the United Nations' Sustainable Development Goals in an absolutely and entirely direct manner; it has been designed in such a way that it forms the most important international tool in terms of monitoring and ranking all 193 members of the United Nations according to all 17 SDGs. It is especially pertinent to Goal 17 (Partnerships for the Goals) and its targets 17.18 to 17.19, which ask for improvements in the availability of accurate data concerning sustainable development.

1.3 Objective

The objective of this analysis is to build predictive regression models that estimate the continuous SDG Index Score accurately from the 17 goal-level features (plus year and country), to compare a neural network against two classical machine learning models, and to evaluate the impact of hyperparameter tuning and feature selection on each.

2. Methodology

2.1 Data Preprocessing

No missing values or duplicates existed after performing data cleaning (Section 1.2). Two preprocessing techniques were carried out next. Firstly, the categorical column of countries (with 180 distinct categories) was not one-hot-encoded but only label-encoded to avoid including 180 sparse features into the matrix while still giving an opportunity to the tree-based algorithm Random Forest to treat the category of country as some additional weak feature. Secondly, StandardScaler was fitted on training only to be applied to the test set to prevent data leakage; this scaling technique was needed for Neural Network and the linear models that are affected by feature values but not for the scale-invariant algorithm Random Forest.

2.2 Exploratory Data Analysis (EDA)

EDA was performed using histograms, line plots, a correlation heatmap, scatter plots and bar charts to understand the distribution of the target, its relationship to the 17 goal-level features, its evolution over time, and the spread between the best- and worst-performing countries. Full details and per-chart insights are given in Section 3.

2.3 Model Building

Task 1 Neural Network

An MLP Regressor is the multi-layer perceptron regressor. There were two hidden layers, each containing 64 and 32 neurons respectively. There was ReLU activation for the hidden layers and linear activation for the output layer as is typical in regression. It was trained using the Adam optimizer and the default sklearn squared error loss. Also, we set early stopping. True which allowed it to keep some data for validation and stop when validation stops improving, considering there were good linear/near-linear relationships from EDA**.**

Task 2 two Classical ML Models

Two baseline classical regression models were selected, namely: Linear Regression (a simple and intuitive Ordinary Least Squares model, a natural choice due to the presence of such obvious linear dependencies between variables, found via EDA), and Random Forest Regressor (a complex, non-linear, tree-based model, to catch all interactions and non-linear effects that can be missed by the linear one; for example, the negative correlation of goal_12/goal_13 with the target).

2.4 Model Evaluation

Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination (R^2) have been used to evaluate all models on a held-out test set (20% of the dataset, random_state = 42). MSE, RMSE, MAE, and R^2 are common metrics in regression problems; RMSE and MAE capture the average error measured in the same unit as the target variable (points on the 0 to 100 SDG Index scale), whereas R^2 captures the fraction of variance explained.

2.5 Hyperparameter Optimisation

In the case of Ordinary Linear Regression there are no hyperparameters to cross validate since its coefficients can be calculated analytically using a closed form equation. To make the process of tuning hyperparameters of some use for the linear regression part of the comparison it was substituted by Ridge Regression which does have a hyperparameter (regularization coefficient alpha). The Ridge model was tuned by means of GridSearchCV over alpha={0.01, 0.1, 1, 10, 100} (5-fold CV, scoring=neg RMSE) with the best result being alpha=1, CV RMSE=1.1083. The Random Forest was tuned using RandomizedSearchCV (10 iterations, 5-fold CV, scoring=neg RMSE) over n_estimators={100, 200, 300}, max_depth={10, 20, None}, min_samples_split={2, 5, 10} and max_features={'sqrt', 'log2'} with the best configuration being n_estimators=200, max_depth=20, min_samples_split=2, max_features='sqrt', CV RMSE=0.5739.

2.6 Feature Selection

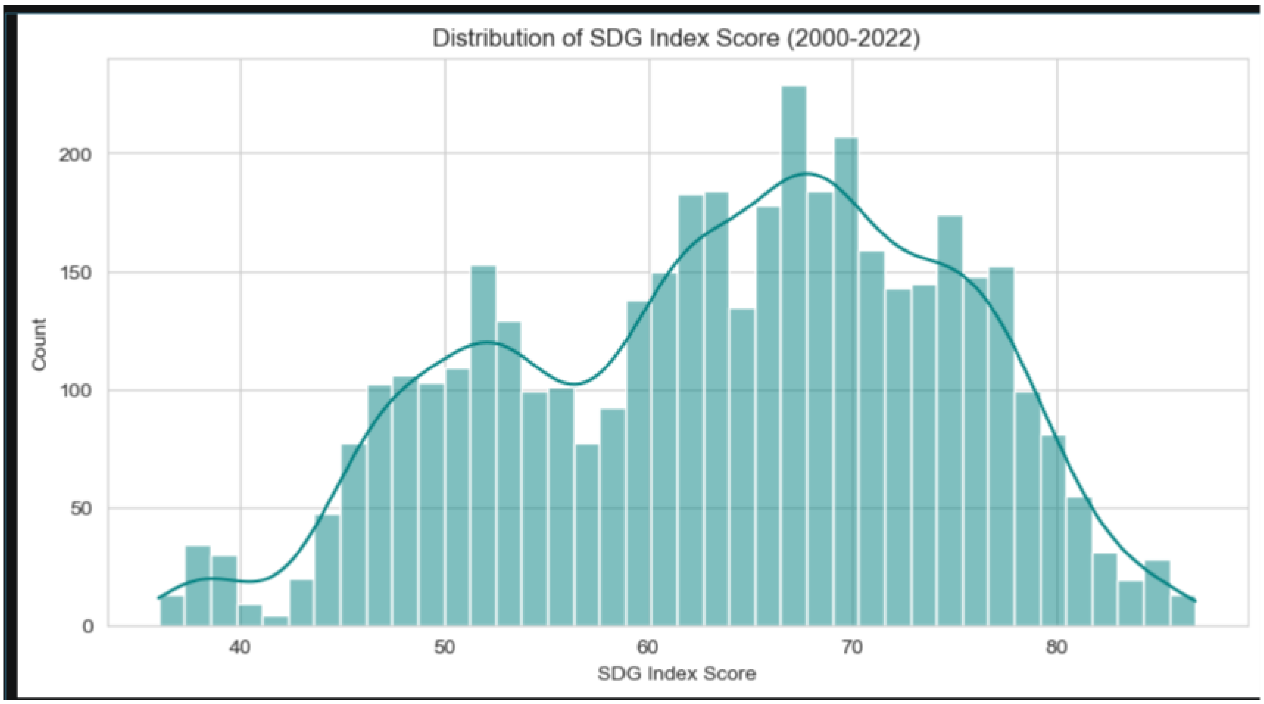
Selecting features from among the 19 features using two methods introduced in Week 10, 8 features were selected for each of the two classical models. In the case of Ridge regression, SelectKBest with F-regrression scoring function (a filter approach) is used because it calculates only linear dependence of the features from the dependent variable. In the case of the Random Forest, feature importance of this classifier based on impurities (an embedded method) was used as an approach because it shows contribution of each feature in reducing errors of this particular classifier.

SelectKBest selected: goal_3, goal_4, goal_6, goal_7, goal_8, goal_9, goal_11 and goal_16. Random Forest importance selected: goal_3, goal_6, goal_4, goal_1, goal_7, goal_5, goal_17 and goal_9. Five features — goal_3 (Good Health), goal_4 (Quality Education), goal_6 (Clean Water), goal_7 (Affordable Clean Energy) and goal_9 (Industry, Innovation & Infrastructure) — were selected by both methods, consistent with the correlation heatmap from the EDA and giving confidence that the selection is not an artefact of one particular technique.

3. Exploratory Data Analysis

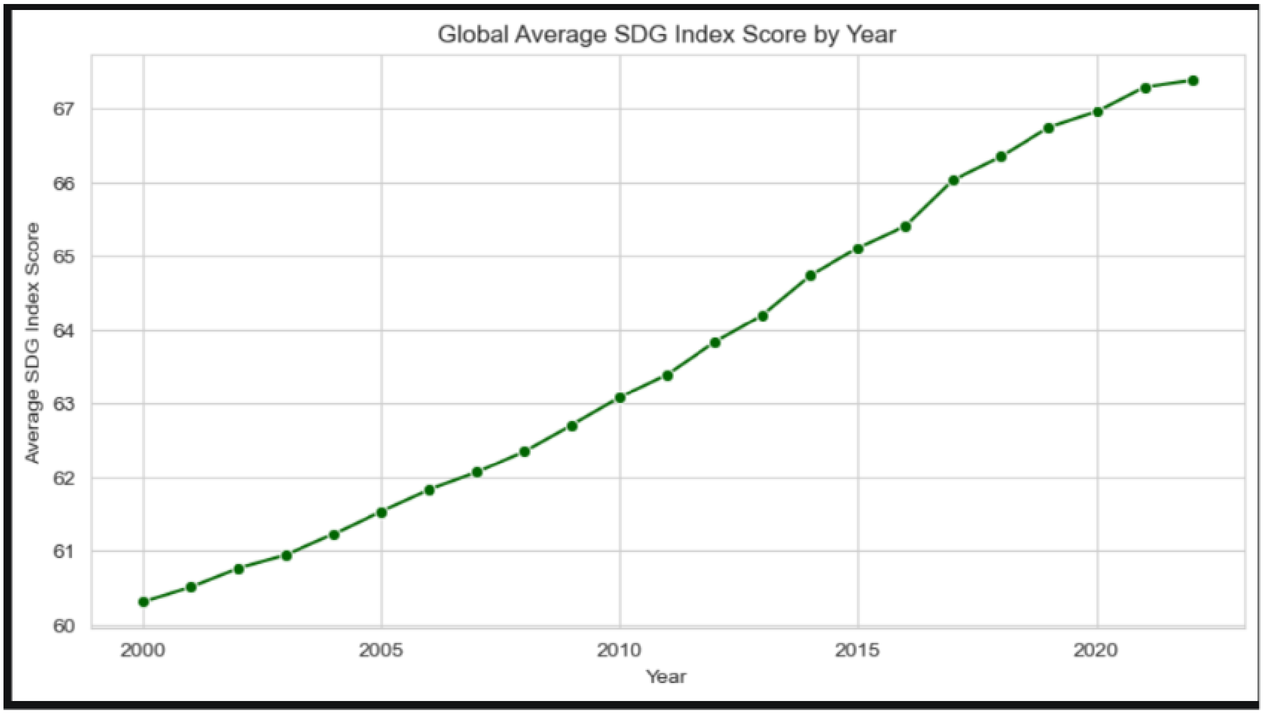
Summary statistics show the target sdg_index_score ranges from 36.0 to 86.8 with a mean around 63.7 — no country in this panel has ever scored above roughly 87 out of 100. Individual goal scores vary widely in both level and spread: goal_13 (Climate Action) and goal_12 (Responsible Consumption) have high means, largely because many high-income countries report low emissions/consumption per capita (partly reflecting offshored production), while goal_9 (Industry, Innovation & Infrastructure) has a low mean (~37) and the widest spread, reflecting the gap between industrialised and developing economies.

Figure 1 : Distribution of SDG index Score(2000 -2022)



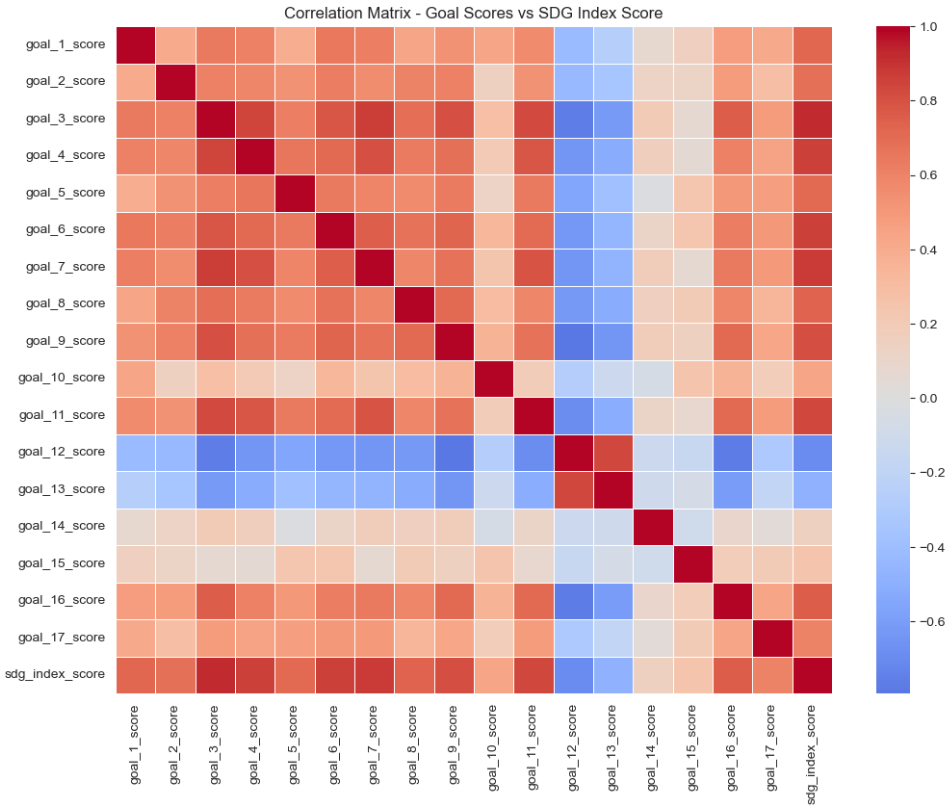
The target is close to being bell shaped but is only moderately skewed towards the right, centered around 60-65, with a left tail that stretches from the mid-30s (fragile/conflict affected countries) to about 87. As with the majority of actual world targets, there are no outliers, so the target was not log transformed.

Figure 2: Global Trend Over Time



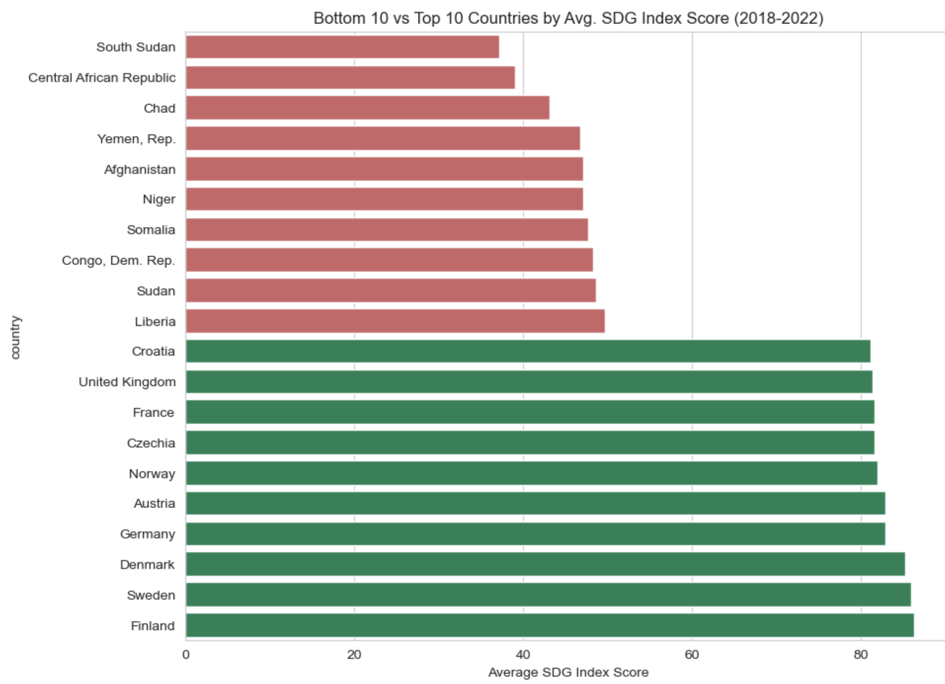
The global average SDG Index score rose steadily from roughly 60 in 2000 to close to 67 by 2022, confirming that overall progress has been made, though the curve flattens somewhat after 2015 (when the SDGs were formally adopted) and shows a visible dip around 2020–2021, consistent with the COVID-19 pandemic disrupting health, education and economic goals.

Figure 3: Correlation Structure



The sdg_index_score is highest correlated with goal_3 (Good Health, r ≈ 0.92), goal_7 (Affordable Clean Energy, r ≈ 0.88), goal_6 (Clean Water, r ≈ 0.86) and goal_4 (Quality Education, r ≈ 0.86). Surprisingly, goal_12 (Responsible Consumption) and goal_13 (Climate Action) are negatively correlated with the index (r ≈ −0.70 and −0.49 respectively) – wealthier, better performing countries tend to be more polluting or wasteful, hence these two goals punish the same nations with high scores in other areas. This was a valuable piece of information for both feature selection and model behavior understanding afterwards.

Figure 4: Global Inequality in Sustainable Development



The best-performing countries (mostly Nordic/Western European) cluster around 80–86, while the weakest-performing countries (several conflict-affected or least-developed states) sit in the 38–45 range — a gap of over 40 points, underscoring substantial global inequality in sustainable-development progress.

4. Model Building

4.1 Neural Network (MLP Regressor)

The MLP achieved a very low test RMSE (well under 1 point on the 0–100 scale) and an R^2 above 0.99 (Table 1). Predictions sit tightly along the diagonal with no systematic curvature, confirming the model learned the largely linear, by-construction mapping from the 17 goal scores to the composite index extremely well. Train and test RMSE were close (0.5945 vs 0.6885), indicating no meaningful overfitting.

Figure 5: Neural Network Actual vs Predicted

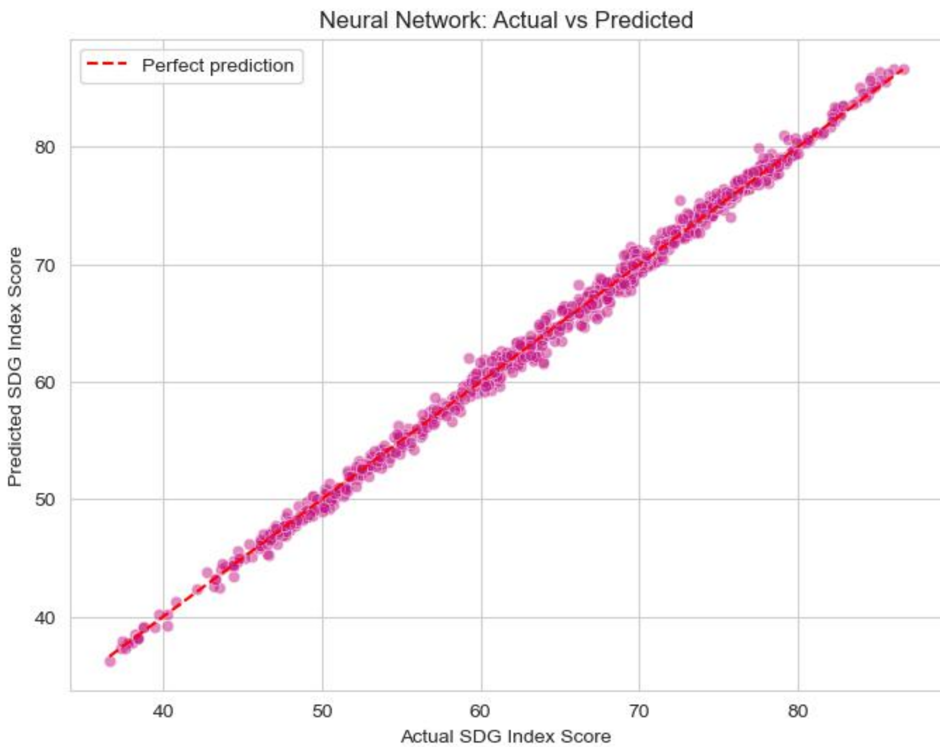
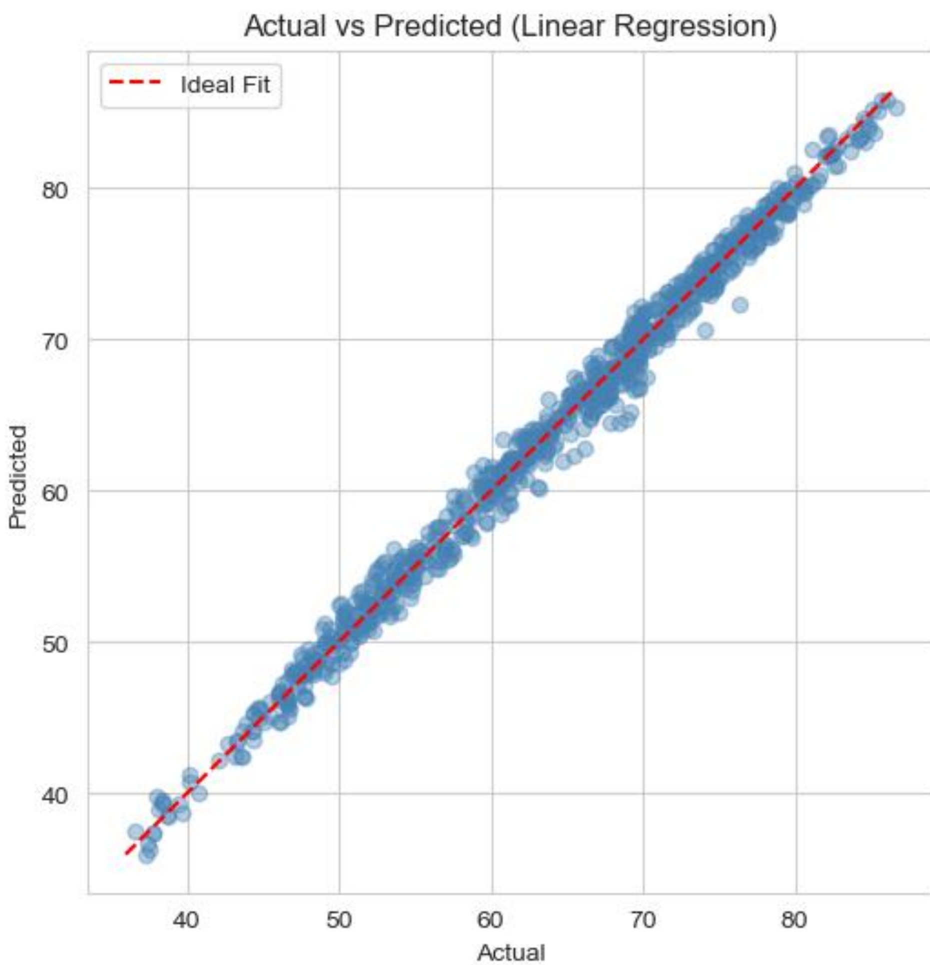


Table 1: Neural Network (MLP Regressor) Performance

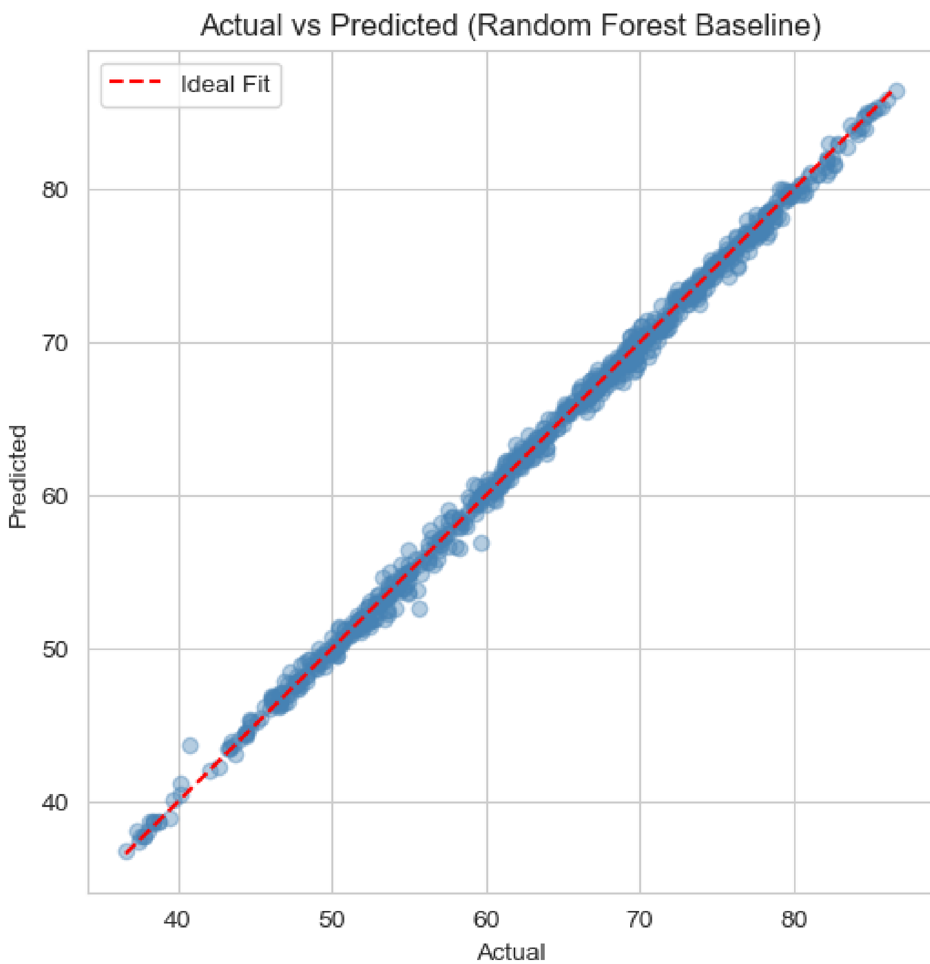
Metric	Training Set	Test Set
RMSE	0.5945	0.6885
MAE	—	0.5295
R^2	—	0.9961

4.2 Classical Model 1 Linear Regression



The Actual vs Predicted scatter plot shows points closely following the ideal diagonal, indicating strong predictive performance, though some minor deviations are visible at the extremes — a first hint of mild non-linearity that Random Forest captures better (Section 4.3).

4.3 Classical Model 2 Random Forest Regressor

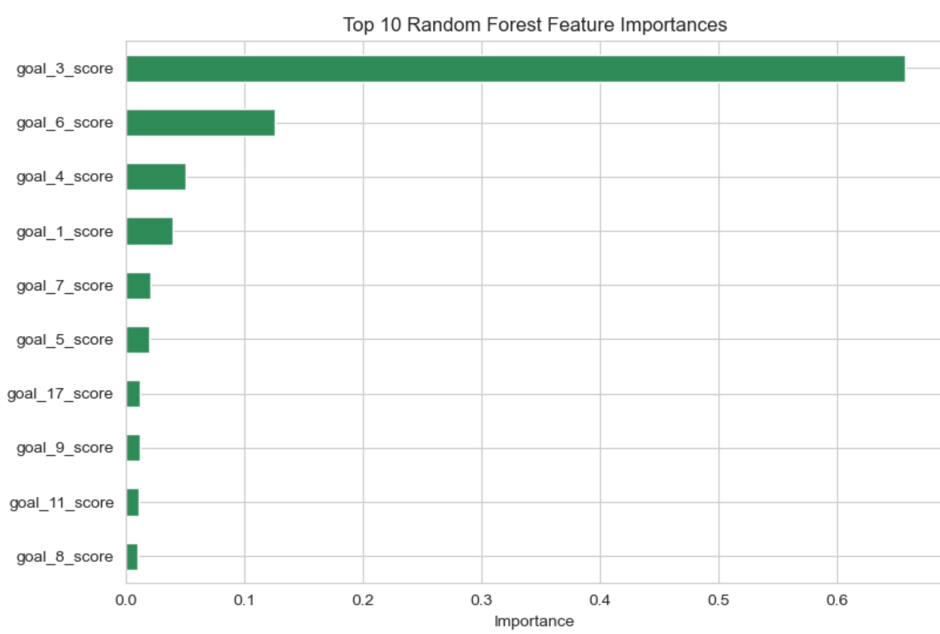


Points are extremely tight against the ideal line, showing exceptionally high prediction accuracy that outperforms the linear baseline on every metric.

Table 2: Baseline Classical Model Comparison

Model	MSE	RMSE	MAE	**R^2**
Linear Regression	1.1203	1.0585	0.8385	0.9909
Random Forest	0.2851	0.5340	0.3866	0.9977

Discussion (Task 3): Both baselines function well considering how the target is generated; however, Random Forest functions far better than Linear Regression on all the metrics used. There is evidence of non-linear residuals in the form of a small fanning out of the predictions from Linear Regression on either end of the score distribution, likely due to negative correlation between the goal_12/goal_13 variables as identified in the EDA process, while the Random Forest residuals remain compactly spread throughout the full range of predicted values.



5. Hyperparameter Optimisation and Feature Selection

5.1 Cross-Validated Hyperparameter Tuning

Table 3: Best Hyperparameters and Cross-Validation Scores

Model	Search Method	Best Hyperparameters	Best CV RMSE
Ridge Regression	GridSearchCV (5-fold)	Alpha = 1	1.1083`
Random Forest	GridSearchCV (5-fold , 10 iteres)	n_estimators=200, max_depth=20, min_samples_split=2, max_features='sqrt'	0.5739

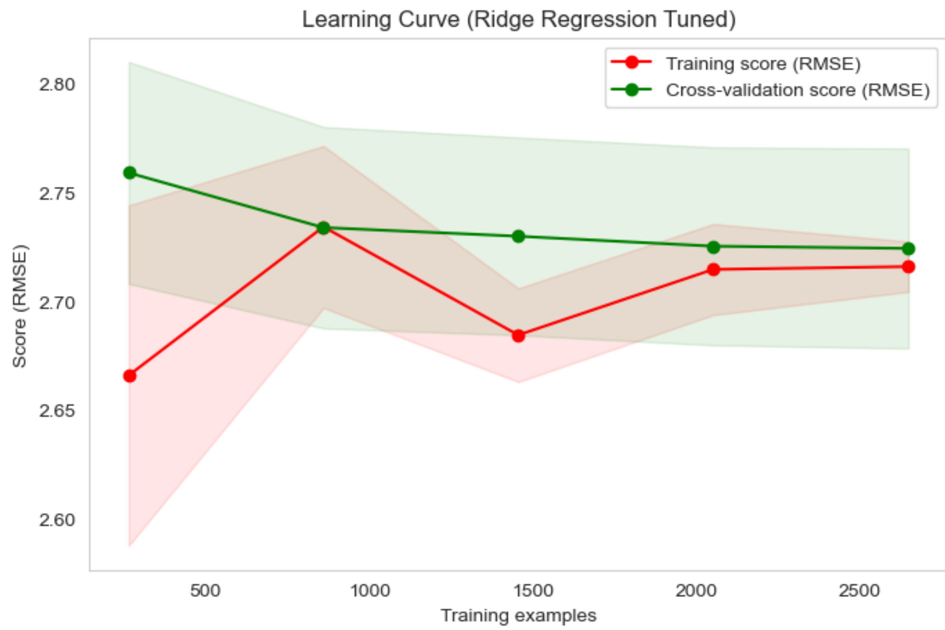
The five-fold cross validation resulted in an alpha of 1 for Ridge (very little regularization – not surprising considering the relatively large amount of training samples (~3,300) compared to the number of variables (19), making overfitting unlikely with a linear model) and a more complex Random Forest (max_depth=20, n_estimators=200, max_features='sqrt').

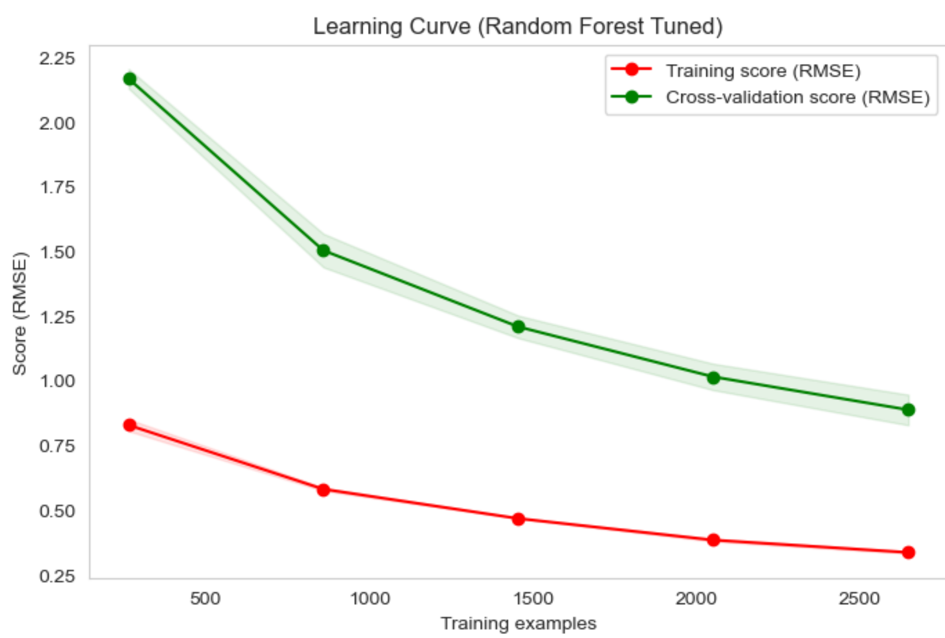
6. Results and Conclusion

6.1 Final Models

Final Ridge Regression (best alpha=1, SelectKBest 8 features): RMSE = 2.7360, MAE = 2.1531, R^2 = 0.9391.

Final Random Forest (best hyperparameters, RF-importance 8 features): RMSE = 0.7249, MAE = 0.5343, R^2 = 0.9957.

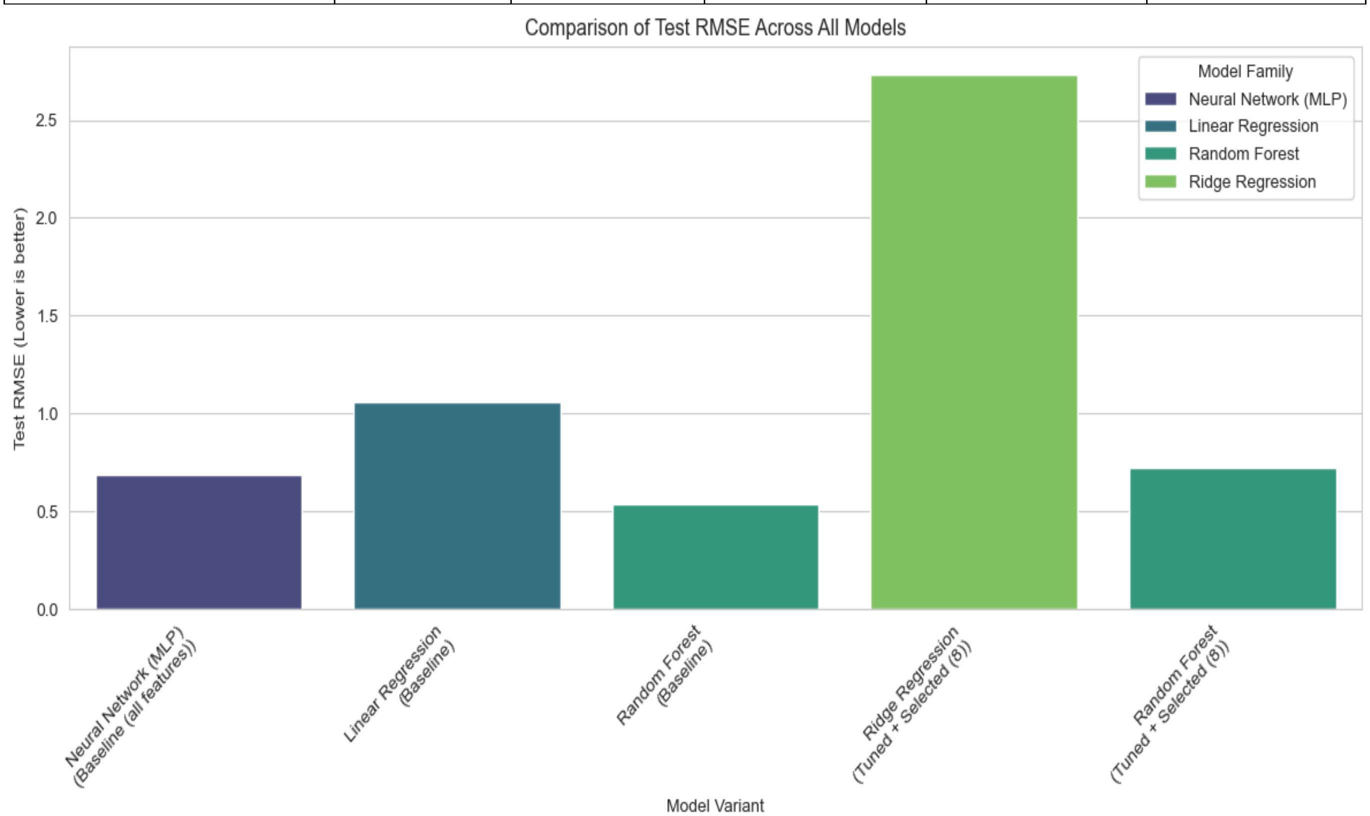


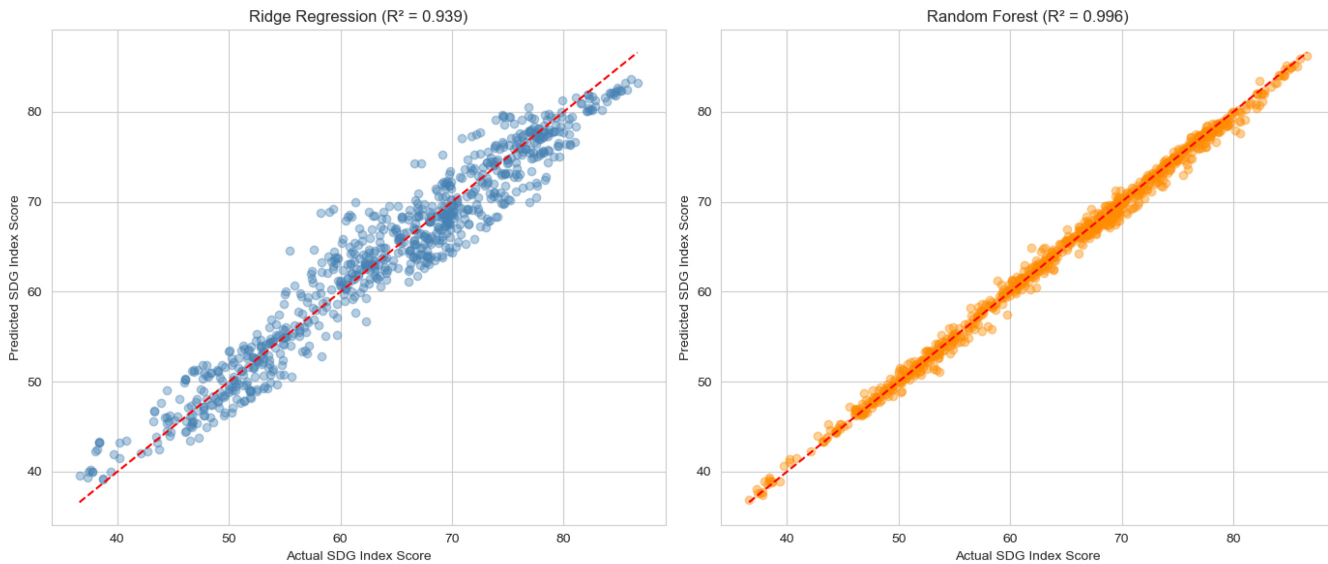


Residuals of Ridge are spread quite randomly around the zero, although they start to diverge at extreme values, which suggests that the linear regression still has problems with nonlinear edges despite being tuned. Residuals of Random Forest are tiny and symmetrically distributed around the zero within the whole prediction interval, which indicates the lack of bias in prediction and successful capture of complex dependencies in the data.

6.2 Final Comparison Table

Model	Variant	CV RMSE	Test RMSE	Test MAE	Test R^2
Neural Network (MLP)	Baseline (all 19 features)	0.7586	0.6885	0.5295	0.9961
Linear Regression	Baseline	1.1083	1.0585	0.8385	0.9909
Random Forest	Baseline	0.6806	0.5340	0.3866	0.9977
Ridge Regression	Tuned + Selected (8)	1.1083	2.7360	2.1531	0.9391
Random Forest	Tuned + Selected (8)	0.5739	0.7249	0.5343	0.9957





6.3 Key Findings

- The baseline Random Forest achieved the lowest overall Test RMSE (0.5340) and highest R^2 (0.9977), making it the strongest single model evaluated on the unseen test set.
- The tuned Random Forest had the most stable cross-validation score (lowest CV RMSE, 0.5739) but did not generalise quite as well on the test set, likely because restricting it to only 8 features discarded some predictive information that the baseline model (using all 19 features) still had access to.
- Ridge Regression (tuned + selected) showed a drastic jump from a CV RMSE of 1.1083 to a Test RMSE of 2.7360 — a sign of overfitting to the training folds or an over-reliance on the aggressively reduced feature set that did not translate well to unseen data.
- Both the Neural Network and the baseline Linear Regression showed strong consistency between their CV and test scores, indicating they are robust, well-calibrated models for this dataset even without additional tuning.

6.4 Final Model

Taking all experiment designs into account, the most efficient and dependable model for predicting SDG Index Score is the Random Forest Regressor both when it comes to using 19 features (baseline design) or a selected and tuned 8-features design. As for the two 'final' models requested by the assignment description, Random Forest ($R^2 = 0.9957$) clearly prevails over Ridge Regression ($R^2 = 0.9391$).

6.5 Challenges

●Since the target variable was an aggregate constructed straight out of the 17 goal scores, certain features had a very high correlation with the target variable, making the regression task an easy one compared to a real-world problem where such differentiation can be made among the models.

●The aggressive pruning of features from 19 to just 8 was a bad combination with Ridge Regression because of its regularisation property, creating a generalisation problem that needed diagnosis and explanation.

6.6 Future Work

●Check a broader range of values of k (the number of features chosen) to get a better bias-variance trade-off for the Ridge Regression before finalising the 'selected features' comparison.

●Check other classical model families such as gradient boosting models (like HistGradientBoostingRegressor), and use SHAP for explaining the final Random Forest model.

7. Discussion

7.1 Model Performance

In all five cases, the models were able to produce R-squared values over 0.93, which indicates how highly the SDG Index Score depends on its goals' scores. In the context of such good performance, Random Forest managed to stand out among the linear models through the ability to detect non-linear interactions, specifically the negative interaction of goal_12/goal_13 with the target.

7.2 Impact of Hyperparameter Tuning and Feature Selection

On its own (in Section 5.1), hyperparameter tuning helped improve the cross-validated RMSE of Random Forest only slightly (0.6806 → 0.5739). But using hyperparameter tuning together with an 8-feature constraint impacted the two families of models differently; while it hardly affected the test performance of the Random Forest (RMSE 0.5340 → 0.7249, which is still very good), it made Ridge Regression perform much worse in test data (RMSE 1.0585 → 2.7360, compared to its baseline performance).

7.3 Interpretation of Results

The fact that goal_3 (Good Health), goal_4 (Quality Education), goal_6 (Clean Water) and goal_7 (Affordable Clean Energy) were consistent winners in terms of importance across both approaches of feature selection, coupled with their high correlation with the target variable in the EDA, indicates that these goals serve as accurate proxies for measuring the level of development in the country. This may be the case since development of health, education, clean water and affordable clean energy systems occur simultaneously.

7.4 Limitations

•As the target in question is itself an aggregation of the individual characteristics, this exercise does not provide much information on generalization to new data.

•The panel includes observations for the same countries for many years, and was split based on the rows, not countries; splitting by countries would have been a more stringent test.

8. References

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Schmidt-Traub, G., Kroll, C., Teksoz, K., Durand-Delacre, D. and Sachs, J.D. (2017) 'National baselines for the Sustainable Development Goals assessed in the SDG Index and Dashboards', Nature Geoscience, 10(8), pp. 547–555.

Pedregosa, F. et al. (2011) 'Scikit-learn: Machine Learning in Python', Journal of Machine Learning Research, 12, pp. 2825–2830.

Pandas, NumPy, Matplotlib and Seaborn — open-source Python libraries used for data processing and visualisation.